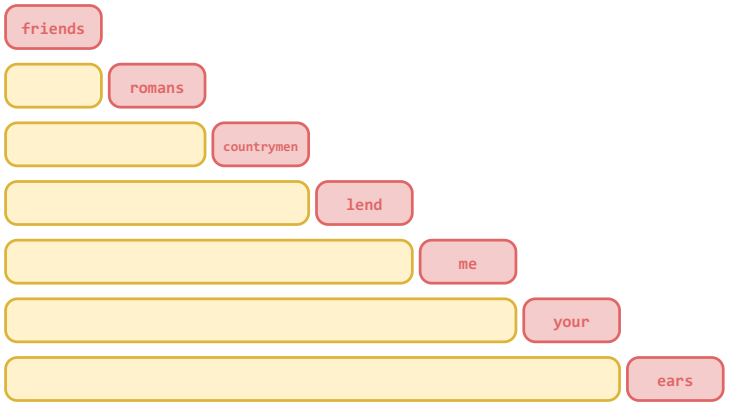
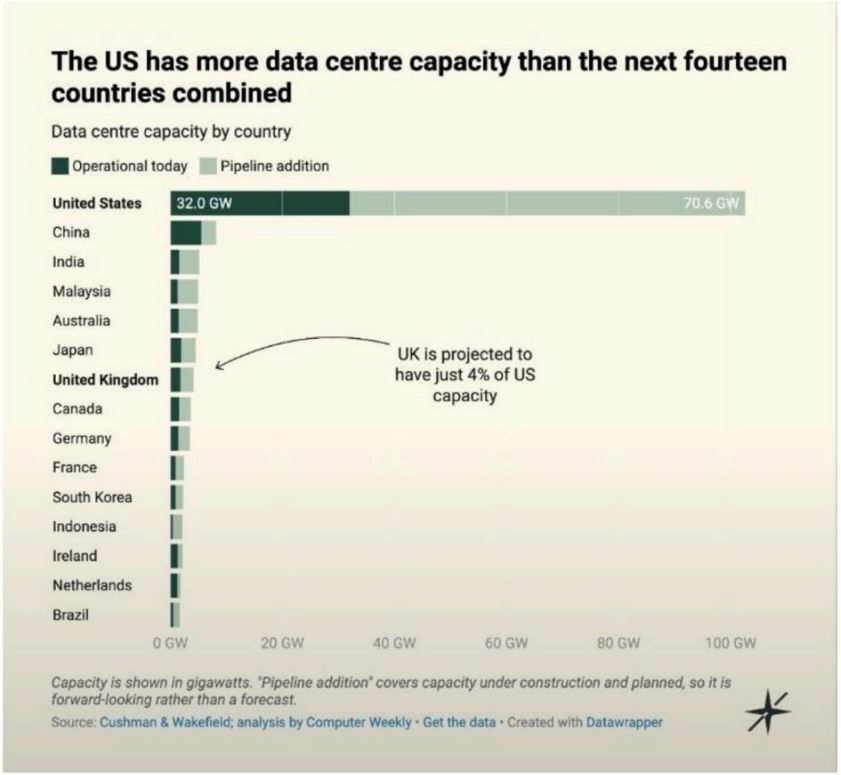
UNIT 01 · CHAPTER 7

# Recurrent Neural Networks

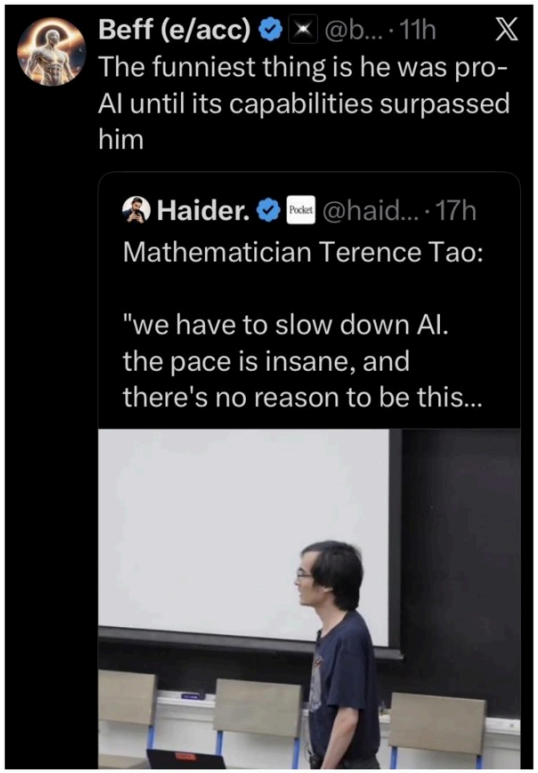



CSE 10124 · Building ChatGPT
William Theisen · University of Notre Dame

# In the News


The US has more data centre capacity than the next fourteen countries combined
Data centre capacity by country
Operational today
Pipeline addition
United States
32.0 GW
70.6 GW
China
India
Malaysia
Australia
Japan
United Kingdom
Canada
Germany
France
South Korea
Indonesia
Ireland
Netherlands
Brazil
UK is projected to have just 4% of US capacity
0 GW
20 GW
40 GW
60 GW
80 GW
100 GW
Capacity is shown in gigawatts. "Pipeline addition" covers capacity under construction and planned, so it is forward-looking rather than a forecast.
Source: Cushman & Wakefield; analysis by Computer Weekly · Get the data · Created with Datawrapper

# In the News

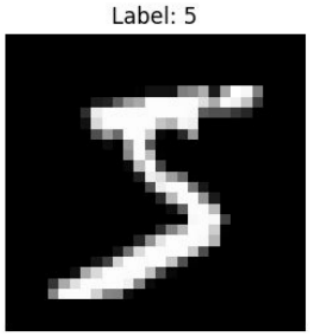

Beff (e/acc) @b... · 11h
The funniest thing is he was pro-AI until its capabilities surpassed him
Haider. @haid... · 17h
Mathematician Terence Tao:
"we have to slow down AI. the pace is insane, and there's no reason to be this...

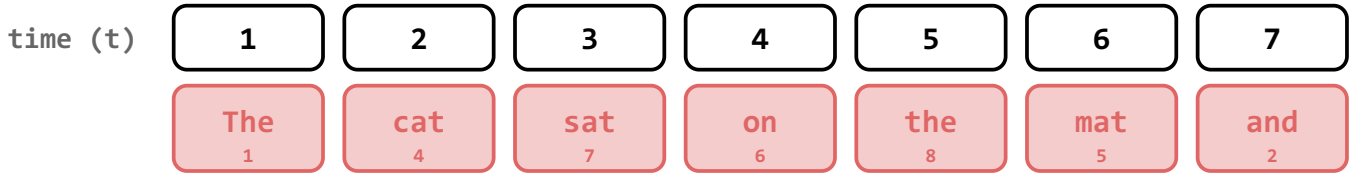

etn.
@etnshow
X.com
JUST IN: OpenAI are offering robotics engineers salaries up to $500,000 before equity as they push into hardware, per BI.
05:19 · 9/18/26 · 739K Views
14
120
1.7K
547


MTS @MTSlive · 1h
SITUATION DETECTED:
Anthropic has set up a wet lab in the San Francisco Bay Area for physical biology work. It wants to unlock treatments for rare diseases, and its research has now gone beyond in silico evaluations, per Reuters.
43
81
963
41K

# Reading vs. Seeing

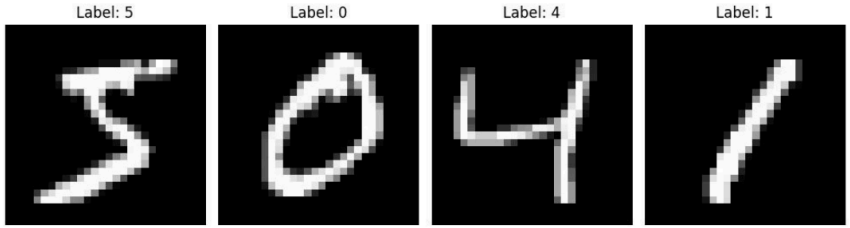
When we **see** something, we generally take it in all at once.



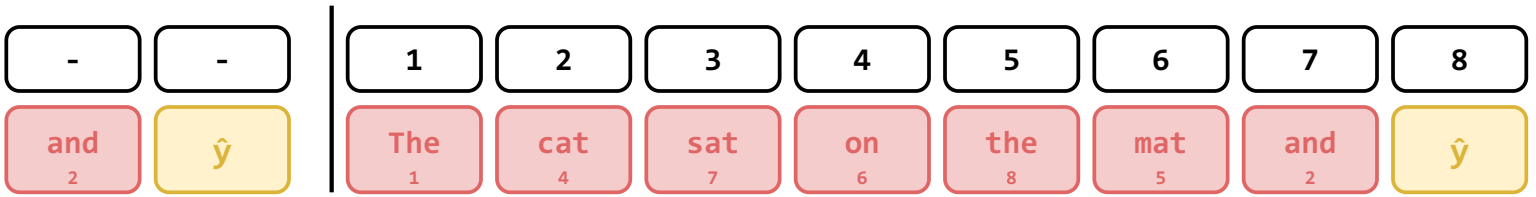

However, when we **read**, we typically read in a sequence, one word at a time. We could say that reading has a dimension of `time (t)`.

# Reading vs. Seeing




When we're looking at images, it doesn't really matter what image we saw previously.

However, as we read across `time (t)`, what we've seen previously matters a LOT! It **provides context to the token we're currently seeing**.



# Prediction in the Brain

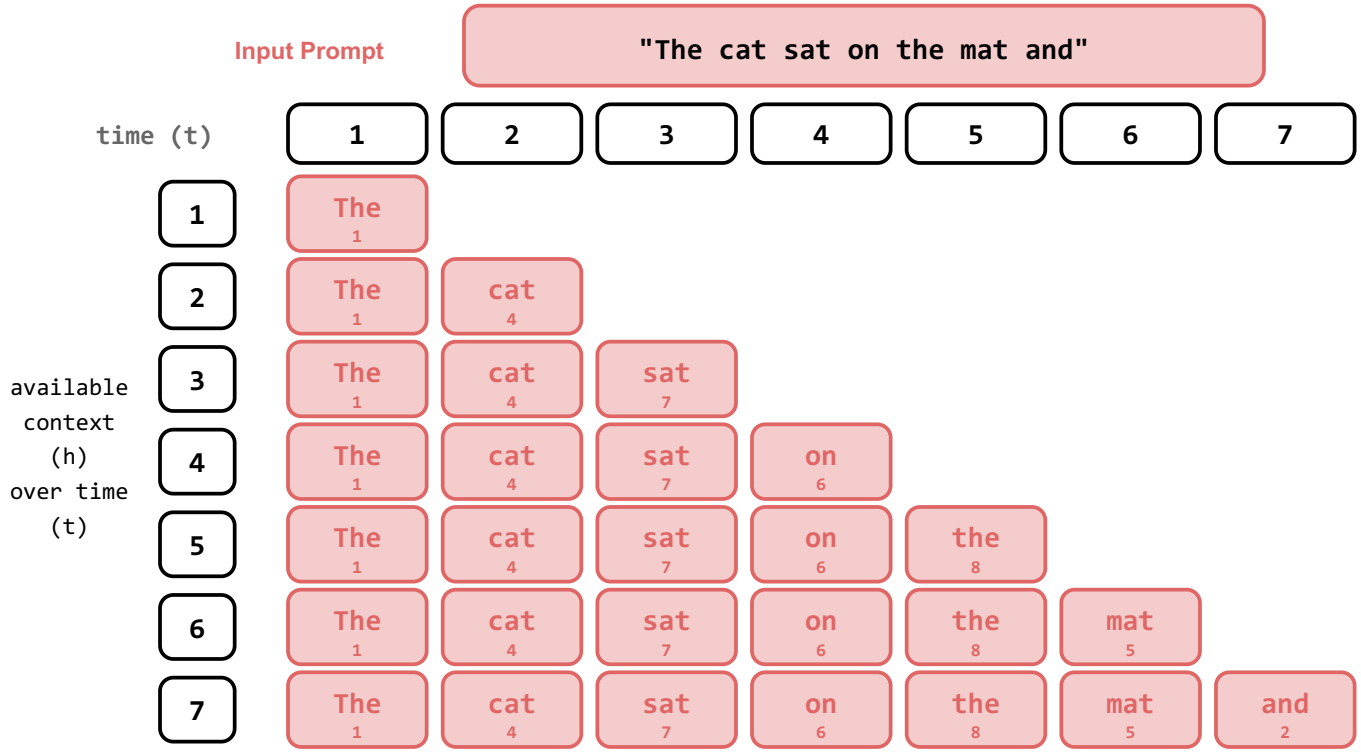
**Goldstein et al.** (*Nature Neuroscience*, 2022) played a 30-minute podcast (5,113 words) and stopped before every word to ask people to guess it.

- People guessed the next word right 28% of the time on average.
- GPT-2's guesses lined up with theirs (correlation 0.79), and lined up better the more of the story GPT-2 could see, up to about 100 words back.
- Electrodes on the brains of 9 epilepsy patients picked up signs of the **upcoming word hundreds of milliseconds before it was heard**, and a **surprise** signal after a word they didn't expect.

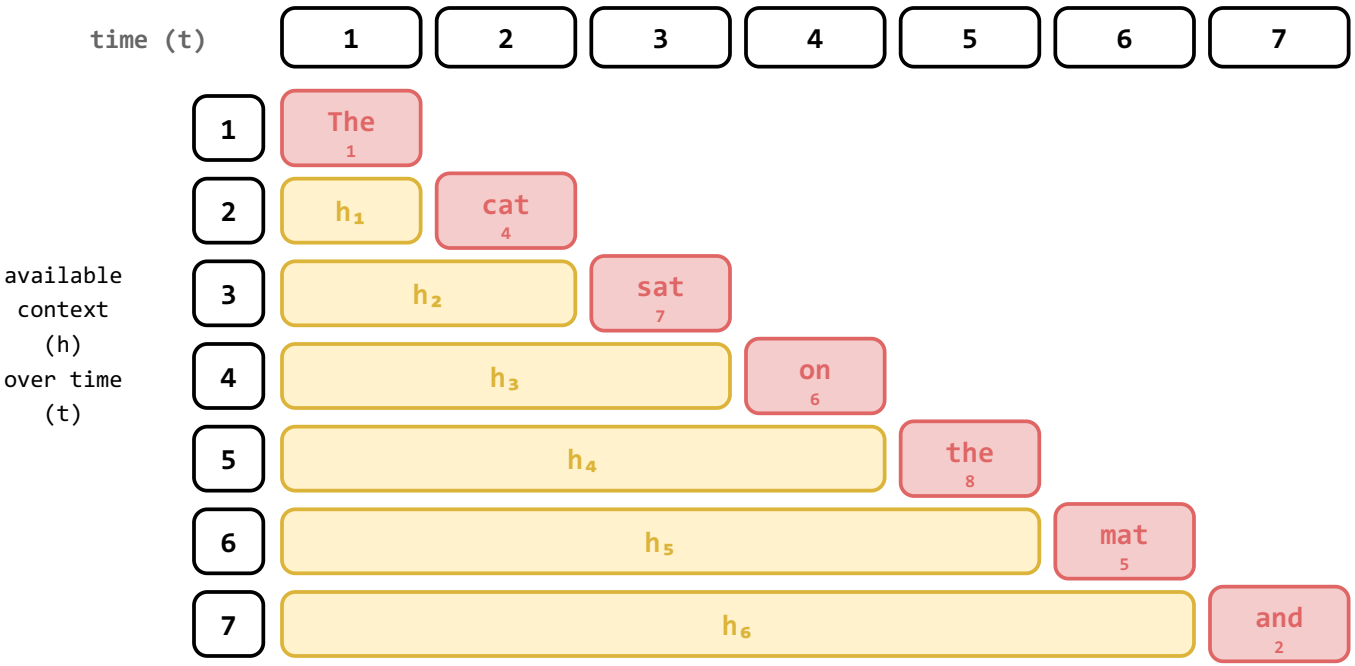
Our brains also predict the next word, and they need the context to do it.

**Careful:** matching a model's guesses says nothing about how the brain makes them. The brain is not an RNN.

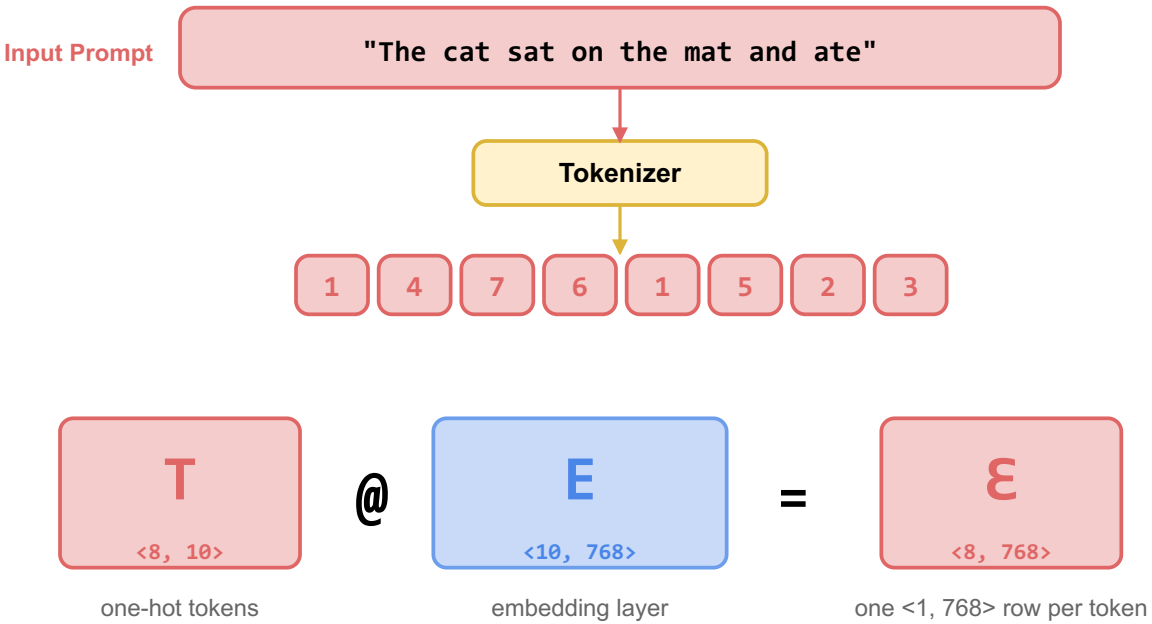
# Time and Context



# Time and Context



# Producing Text Embeddings for an input prompt



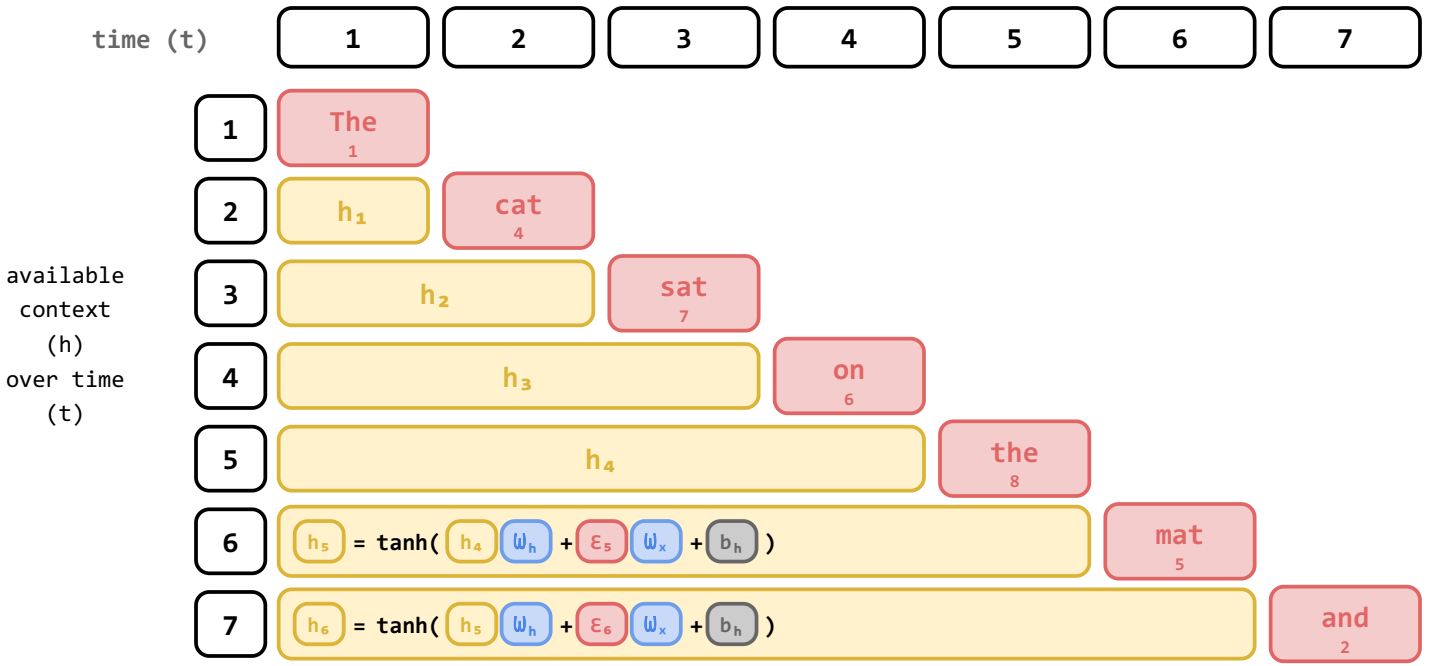
# Remembering Context

As we read through our **input prompt** over `time (t)` we need to **remember**, **maintain**, and **build up the context** of the sequence of **embeddings** we've seen so far.

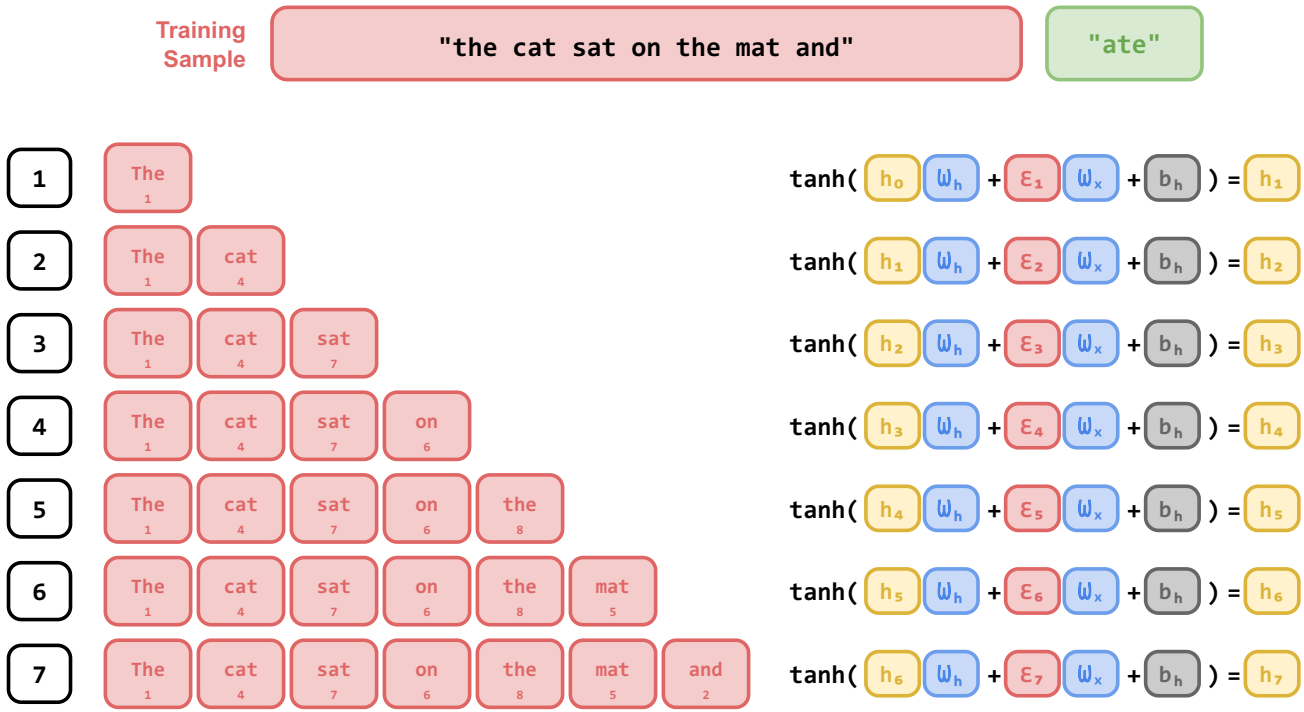
$$[7.1] \quad h_t = \tanh(h_{t-1} W_h + \varepsilon_t W_x + b_h)$$

- $h_t$ = context at `time t` (current timestep)
- $h_{t-1}$ = context from `time t-1` (prior timestep)
- $\varepsilon_t$ = **embedding** for current **token**

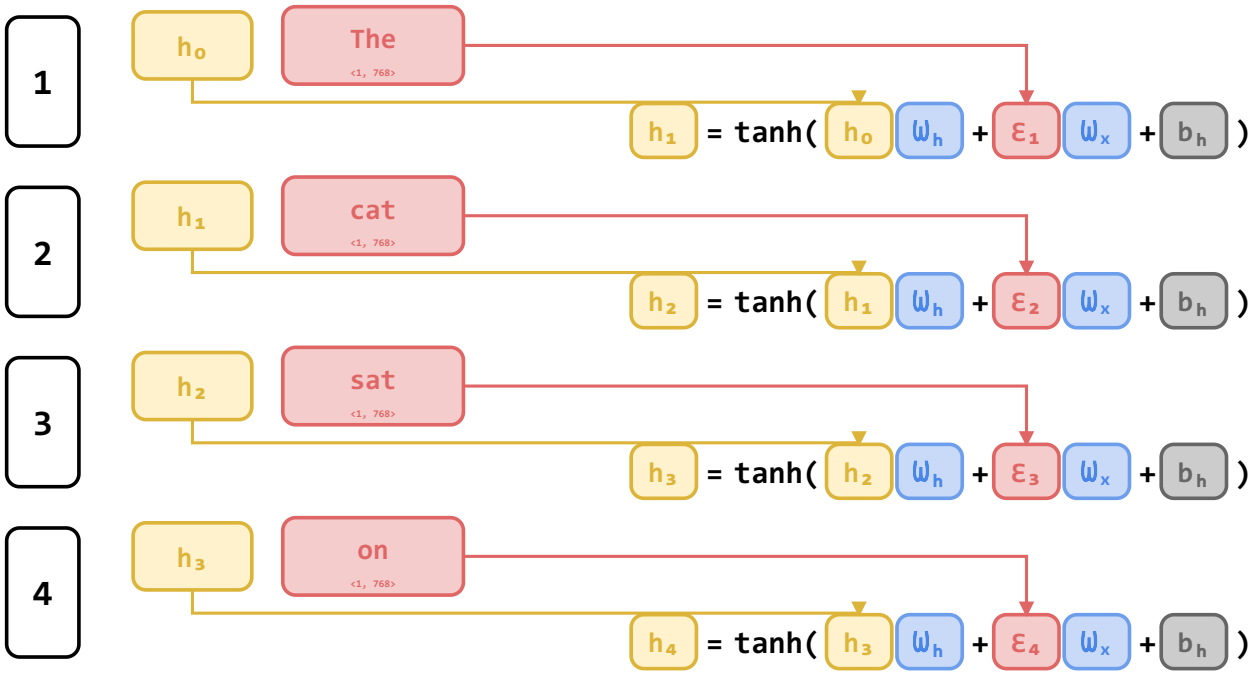
# Time and Context



# Building Context over time (t)



# Building Context over time (t)



# Producing output at time (t)

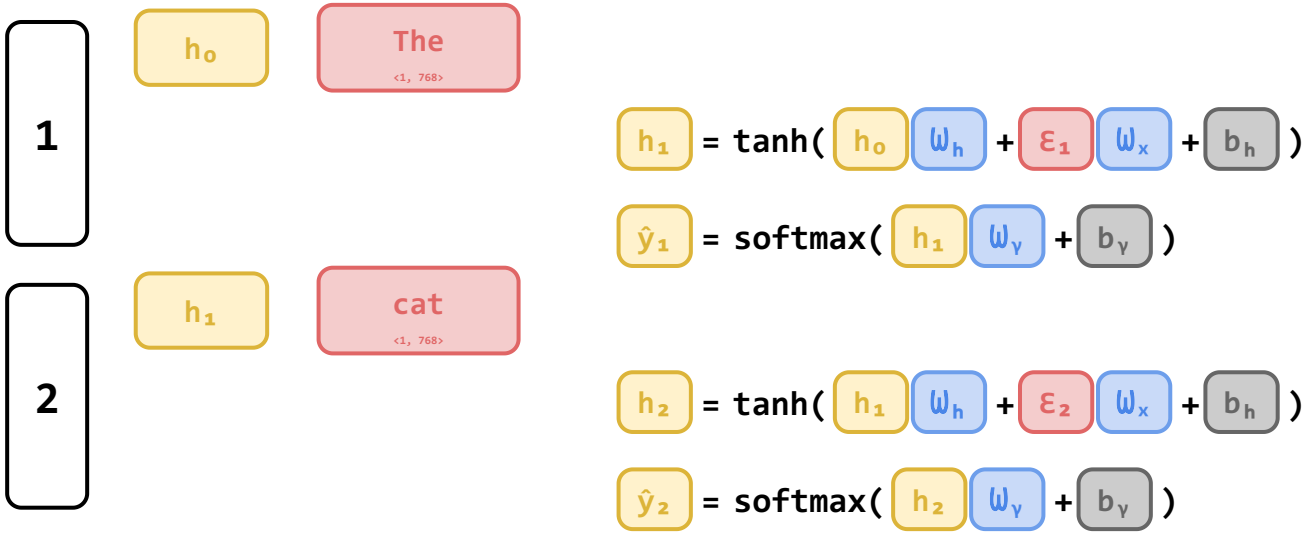
$$h_7 = \tanh(h_6 W_h + \varepsilon_7 W_x + b_h)$$

$$\hat{y} = \text{softmax}(h_7 W_y + b_y)$$

| Token ID | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Token | "." | "The" | " and" | " ate" | " cat" | " mat" | " on" | " sat" | " the" |
| $\hat{y}$ | 0.05 | 0.07 | 0.03 | 0.08 | 0.12 | 0.10 | 0.10 | **0.38** | 0.07 |

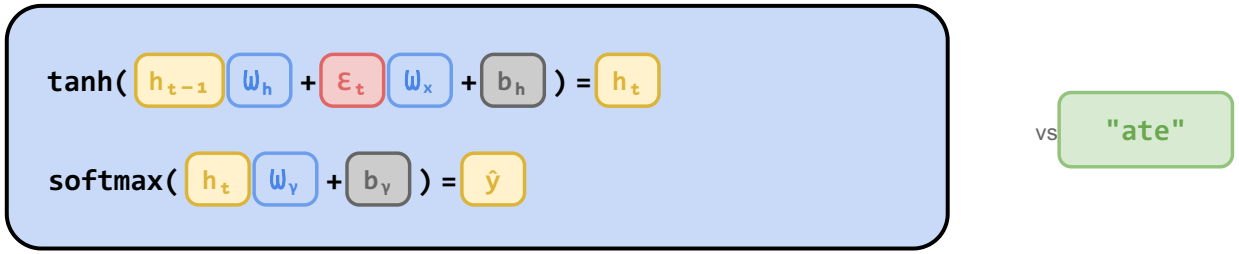
The RNN's most likely next token is token 7, " sat".

# Producing output over time (t)



# Training Recurrent Neural Networks

**Training Sample:** "the cat sat on the mat and", next token "ate"




| Token ID | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Token | "." | "The" | " and" | " ate" | " cat" | " mat" | " on" | " sat" | " the" |
| $\hat{y}$ | 0.05 | 0.07 | 0.03 | 0.08 | 0.12 | 0.10 | 0.10 | 0.38 | 0.07 |
| y | 0.00 | 0.00 | 0.00 | **1.00** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

# Training a Text Prediction Model

